Exp No: 9	Implementation of SUBNETTING in CISCO Packet Tracer Simulator

AIM:

To design two subnets with static IP addresses, connect the subnets through a router in Cisco Packet Tracer and verify the communication between the subnets using the ping command.

ALGORITHM / PROCEDURE:

1. Decide the two networks to be used: Subnet A **192.168.1.0/24** and Subnet B **192.168.2.0/24** (subnet mask 255.255.255.0).
2. Open Packet Tracer and place one router, two switches and the PCs. Connect the PCs to their switch and each switch to the router with copper straight-through cables.
3. Assign a static IP address, subnet mask and default gateway to every PC under *Desktop > IP Configuration*. The gateway of each PC is the router interface of its own network (192.168.1.1 or 192.168.2.1).
4. Open the router CLI and configure the interfaces: `interface gigabitEthernet 0/0`, `ip address 192.168.1.1 255.255.255.0`, `no shutdown`.
5. Repeat step 4 for `gigabitEthernet 0/1` with the address 192.168.2.1.
6. From the command prompt of a PC use `ping` to test the connectivity with a PC of the other subnet.

Router configuration:

Router CLI

```
Router>enable
Router#configure terminal
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
```

Device	Interface	IP address	Subnet mask
Router0	Gig0/0	192.168.1.1	255.255.255.0
Router0	Gig0/1	192.168.2.1	255.255.255.0

DIAGRAM:

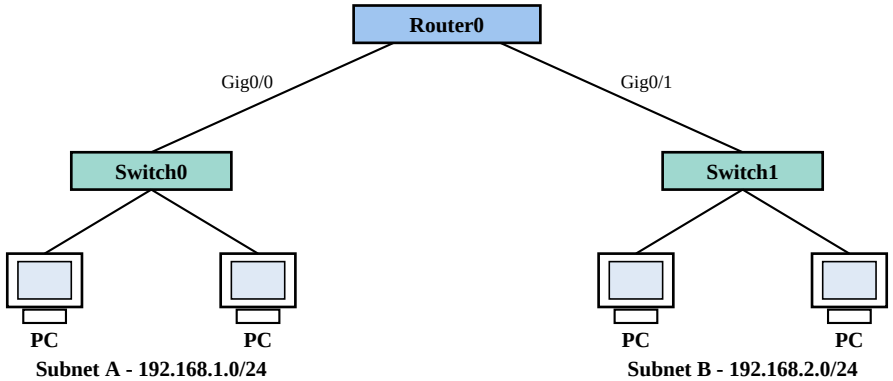
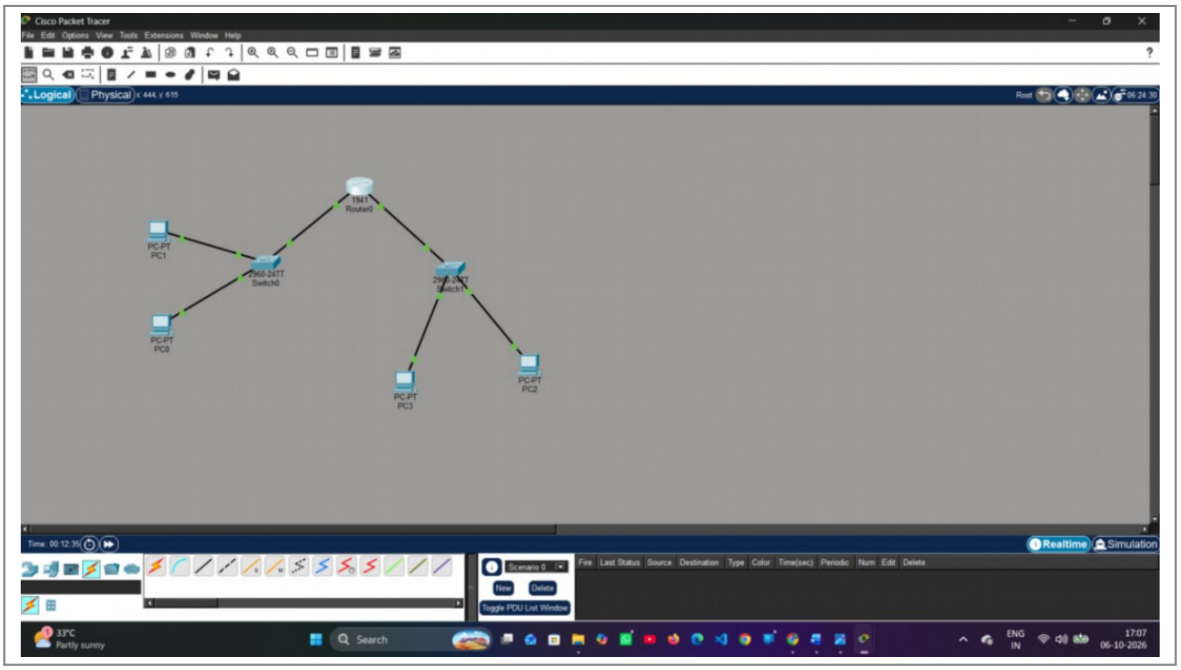


Fig 1: Two subnets connected through a router

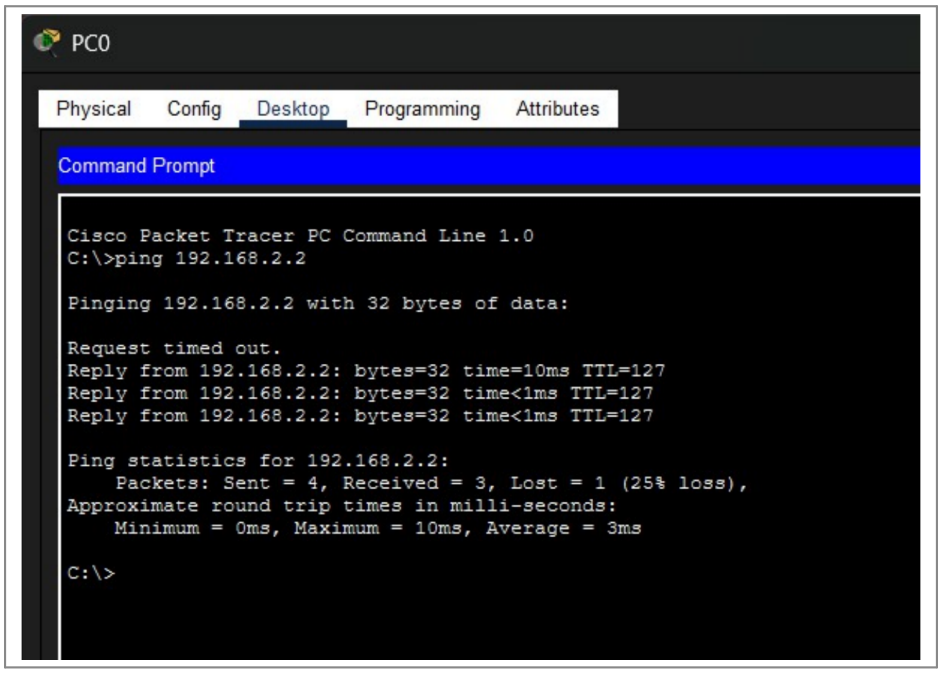
OUTPUT:



Packet Tracer topology - router, two switches and the PCs



Router0 CLI - interface configuration



PC command prompt - ping 192.168.2.2 from the other subnet through the router (3 of 4 replies, TTL = 127)

The first packet is lost (*Request timed out*) because the ARP request has to be resolved first. The remaining replies have TTL = 127, which shows that the packets passed through one router.

RESULT:

Thus two subnets were designed and connected through a router in Cisco Packet Tracer. The hosts of different subnets could communicate with each other through the router.